Supplementary Information

Supplementary Text and Supplementary Figures

for

A FAIR-compliant resource and toxicology knowledge graph for mechanistic exploration of endocrine disrupting chemicals associated toxicity

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Supplementary Text

1. Workflow for the compilation of Endocrine Disrupting Chemicals (EDCs)

The World Health Organization (WHO) and the United Nations Environment Programme (UNEP) report titled 'State of the Science of Endocrine Disrupting Chemicals 2012' states that 'A potential endocrine disruptor is an exogenous substance or mixture that possesses properties that might be expected to lead to endocrine disruption in an intact organism, or its progeny or (sub) populations' (WHO/UNEP, 2013). In line with this definition, a four-staged workflow was developed to systematically compile and curate potential endocrine disrupting chemicals (EDCs) and their associated endocrine-mediated adverse effects, resulting in the **D**atabase of **E**ndocrine **D**isr**u**pting **C**hemicals and their **T**oxicity profiles (DEDuCT), a comprehensive knowledgebase that systematically compiles evidence from published literature (Karthikeyan et al., 2019, 2021). Importantly, this four-staged workflow was independently assessed by a working group of 18 experts convened by the French Agency for Food, Environmental and Occupational Health and Safety (ANSES) (ANSES, 2021). These experts recommended this approach to identify potential EDCs from published literature based on the WHO definition (ANSES, 2021).

The latest version, DEDuCT version 2.0 (released on October 2, 2020) compiled information on 792 potential EDCs with supporting evidence from 2218 articles (Karthikeyan et al., 2021). Since its release, global efforts in testing and identification of EDCs have generated an additional large corpus of published literature. Therefore, in this study, we present an expanded version, DEDuCT version 3.0 (DEDuCTv3.0), that compiles information on 1043 potential EDCs from 3269 research articles published until January 2025. To achieve this, a systematic four-staged workflow based on extensive manual curation was utilized to curate chemicals and their associated endocrine-mediated effects in humans or rodents from published literature (Figure 1) (Karthikeyan et al., 2019). This four-staged workflow is summarized in the following sections.

1.1. Stage 1: Mining EDC-relevant literature

The first stage involves the mining of EDC-relevant published literature from different sources. In comparison to the previous study, it was observed that the sources for published literature, such as the World Health Organization (WHO) report (WHO/UNEP, 2013), the Endocrine Disruption Exchange (TEDx, https://endocrinedisruption.org/) and

EDCs Databank version 2015 (Montes-Grajales and Olivero-Verbel, 2015), were no longer updated or accessible. Therefore, only PubMed (https://pubmed.ncbi.nlm.nih.gov/) was queried with the following query:

"EDCs" OR "EDC" OR ("endocrine" AND "disrupt") OR ("disrupt" AND "endocrine") OR "endocrine disruptors" OR "endocrine-disruptors" OR "endocrine disruptor" OR "endocrine-disruptor" OR "endocrine disrupters" OR "endocrine-disrupters" OR "endocrine disruption" OR "endocrine-disruption" OR "endocrine disruptive" OR "endocrine-disruptive" OR "endocrine disrupting" OR "endocrine-disrupting" OR "endocrine disrupter"

This search, last conducted on January 21, 2025, was performed without any additional time range filters, and resulted in a list of 29688 published literature, of which 14295 articles had not been screened in the previous two versions of DEDuCT. Thereafter, the titles and abstracts of these 14295 new articles were manually screened by two independent curators, with articles split between them, to verify the relevance of the article, ensuring that the identified keywords were specifically associated with EDCs and their related toxicities. This process resulted in the identification of 9136 new articles at the end of stage 1 (Figure 1).

1.2. Stage 2: Literature filter based on study type and test organism

In stage 2 of the workflow, 9136 new articles filtered from stage 1 were screened to identify studies based on *in vivo* evidence in humans, or *in vivo* experimental evidence in rodents or *in vitro* experimental evidence from human cell-line studies. During this stage, the studies comprising receptor-based binding assays or *in silico* methods to infer endocrine disrupting potential were filtered out as the evidence is insufficient to determine if the chemical can lead to adverse effects based on environmental exposures. Additionally, review articles and correlation-based studies were excluded as they lack experimental evidence directly linking chemical exposure to the observed effects. Through this extensive screening process, 1795 new research articles relevant to EDC-specific experiments in humans or rodents were obtained (Figure 1).

1.3. Stage 3: Compilation of tested chemicals from the filtered research articles

In stage 3 of the workflow, the full texts of the filtered 1795 articles were first obtained for comprehensive screening of tested chemicals. It was observed that full texts were not available for nine articles and were therefore excluded from further analysis.

Thereafter, the two curators screened through the 1786 articles, curated the chemical data, and mapped the chemicals to their two-dimensional (2D) structure identifiers using standard chemical databases such as PubChem (https://pubchem.ncbi.nlm.nih.gov/) and Chemical Abstracts Service (CAS) (https://commonchemistry.cas.org/). This process enabled the consolidation of synonymous chemical names and standardization of chemical representations across studies. Chemicals that lacked standardized structural information were excluded due to insufficient chemical composition data. This resulted in the identification of 637 unique chemicals that had been tested for endocrine disruption in humans or rodents in at least one research article among the 1786 screened articles obtained after stage 2 (Figure 1).

1.4. Stage 4: Identification of EDCs with supporting evidence on systems-level endocrine-mediated perturbations

In stage 4 of the workflow, the two curators independently assessed the tested chemicals for the presence of significant endocrine-associated adverse effects in the published literature. First, chemicals were discarded if they were reported as natural hormones across studies. Next, studies associated with a chemical were filtered out if they solely tested the chemical as part of a mixture, or reported only therapeutic effects without mentioning any associated endocrine-mediated adverse effects Furthermore, studies were filtered out if they solely relied on *in vitro* rodent experiments. Thereafter, the observed adverse effects or endpoints were first checked for significance within the associated literature and then checked for endocrine disruption relevance, such as changes to morphology, physiology, reproduction, growth, development, and lifespan. Finally, a chemical was considered as a potential EDC if it contained at least one significant supporting evidence for endocrine-relevant effects from associated literature. Through this extensive manual curation, 479 chemicals were identified with strong supporting evidence for endocrine disruption from 1051 published articles.

Next, the reported adverse effects associated with these 479 chemicals were manually standardized based on the vocabulary established in earlier versions of DEDuCT (Karthikeyan et al., 2019, 2021), resulting in the compilation of 573 endocrine-mediated endpoints. In earlier versions of DEDuCT, the endocrine-mediated endpoints were additionally classified based on the major biological and physiological processes regulated by the endocrine system, including development, growth, reproduction, metabolism, immunity,

behavior, and cancer (Karthikeyan et al., 2019). According to this classification, the endpoints were grouped into seven systems-level perturbations, namely, Endocrine-mediated cancer (CT), Reproductive (RT), Developmental (DT), Metabolic (MT), Immunological (IT), Neurological (NT), and Hepatic (HT) endocrine-mediated perturbations (Karthikeyan et al., 2019). According to this classification, an endpoint can be assigned to multiple categories where its reported effect involved more than one major biological process. Among 573 endpoints, it was observed that 187 endpoints were not present in earlier versions, and were therefore manually classified into these seven systems-level perturbations based on the major biological processes controlled by the human endocrine system. These unified terms can serve as standard biological vocabulary describing the toxicity profiles of EDCs (Karthikeyan et al., 2019). In addition to the compilation of endocrine-mediated endpoints, the corresponding dosage ranges were also compiled from the supporting literature for these 479 potential EDCs. The dosage units were uniformized, and the dosages were recorded as test and effective dosage for each EDC to enable dose-response measurements. The authors discussed and resolved doubts and disagreements during curation and compiled and verified the final data collected by both curators.

Finally, the compiled data was integrated with DEDuCT version 2.0, resulting in a list of 1043 unique potential EDCs, comprising 796 unique endocrine-mediated endpoints systematically compiled from 3269 published literature.

A chronological analysis of the 3269 articles revealed that nearly half the 3269 articles compiled in DEDuCTv3.0 were published between 2017 and January of 2025 (Figure S1c). Notably, it was observed that the reliance on *in vitro* human experiments is gradually increasing over the years, suggesting the reliance on non-animal approaches towards EDC research (Figure S1c). In addition, the literature data was utilized to understand the identification of new EDCs over the years. Here, the supporting evidence for each EDC was sorted chronologically, and the year of identification was defined as the date of the earliest published literature. Through this analysis, it was observed that the number of new EDCs identified has increased over the past few decades, aligning with the growth in research and interest regarding endocrine disruptors (Figure S1d).

2. Additional information on EDCs in DEDuCTv3.0

In addition to experimental evidence, DEDuCTv3.0 also compiles diverse information for the 1043 potential EDCs including two-dimensional (2D) and three-dimensional (3D)

chemical structures, physicochemical properties, predicted ADMET properties, molecular descriptors, associated gene, phenotype and disease information curated from Comparative Toxicogenomics Database (CTD) (Davis et al., 2025) (https://ctdbase.org), active readouts from ToxCast invitrodb v4.3, associated Adverse Outcome Pathways (AOPs) from AOP-Wiki (Table S8), consumer products presence information from CPDat v4, and presence in various regulation lists and in human biospecimens.

The potential EDCs are classified based on their environmental sources into 7 broad categories and 48 sub-categories (Table S1). Further, ClassyFire (Djoumbou Feunang et al., 2016) was utilized to hierarchically classify the EDCs based on their structure (Table S1). In case ClassyFire (Djoumbou Feunang et al., 2016) failed to classify the chemical, the structural definitions were relied upon to obtain the Kingdom of such chemicals (Djoumbou Feunang et al., 2016). Moreover, the 1043 potential EDCs were classified into 4 categories (I-IV) based on the type of supporting evidence for endocrine disruption in published experiments specific to humans or rodents (Table S1). The compiled information in DEDuCT can be downloaded in flat file format from the associated webserver at: https://cb.imsc.res.in/deduct/download.

In sum, we provide an expanded list of 1043 potential EDCs in DEDuCTv3.0, which could assist academia, industry, and regulatory agencies to develop safer consumer products.

3. Exploration of chemical space within DEDuCTv3.0

The chemical space of 1043 EDCs within DEDuCTv3.0 was explored using a chemical similarity network (CSN) and a target similarity network (TSN). To construct the CSN, 1043 EDCs were used as nodes, and the edges denoted structural similarity between corresponding nodes. Based on the previous study, the chemical similarities were computed between chemicals as the Tanimoto coefficient based on their ECFP4 fingerprints (Bajusz et al., 2015; Karthikeyan et al., 2019). Thereafter, the chemical pairs having Tanimoto coefficients > 0.5 were considered as being structurally similar, and were used as edge weights in the network (Figure S3).

To construct TSN, the chemical-gene interaction information was curated from ToxCast invitrodb v4.3 (US EPA, 2025), as explained in section 7.4 titled 'Identification of KEs using ToxCast'. This resulted in the identification of 579 of 1043 EDCs targeting 400 human or rodent relevant genes. Based on the previous study (Karthikeyan et al., 2019), the

similarities between chemicals were computed as the Jaccard index (Jaccard, 1912) of the corresponding target gene sets. Thereafter, the chemical pairs having a Jaccard index > 0.5 were considered as being target-wise similar, and were used as edge weights in the network (Figure S5).

Based on the previous study (Karthikeyan et al., 2019), the correlation between structural similarity and target similarity was further probed for 579 of 1043 chemicals with target gene information from ToxCast (Figure S6). The structural similarity based on the Tanimoto coefficient was plotted on the y-axis, and the target similarity based on the Jaccard Index was plotted on the x-axis. Thereafter, Pearson correlation coefficient was computed to check for correlation between structural and target similarities.

4. Compilation of AOPs within AOP-Wiki

Adverse Outcome Pathway (AOP) is a toxicological knowledge framework that captures biological processes underlying stressor-induced toxicity (Ankley et al., 2010). In this framework, toxicity-related biological events, referred to as Key Events (KEs), are organized sequentially, beginning with the interaction of the stressor with a biological target, known as the Molecular Initiating Event (MIE), and culminating in an Adverse Outcome (AO) (Ankley et al., 2010; OECD, 2017; Villeneuve et al., 2014a, 2014b). The causal, directional relationships between these events are termed Key Event Relationships (KERs) (Ankley et al., 2010; OECD, 2017; Villeneuve et al., 2014a, 2014b). AOPs are stressor-agnostic and context-specific, meaning each AOP is tailored to a particular biological scenario (Ankley et al., 2010; OECD, 2017; Villeneuve et al., 2014a, 2014b). Developed globally, these AOPs are deposited in the AOP-Wiki (https://aopwiki.org), which is the largest, publicly accessible repository hosted by the Society for the Advancement of Adverse Outcome Pathways (SAAOP). AOP-Wiki hosts several AOPs, where each AOP is documented in the form of KEs (including MIE and AO) and KERs, all of which are supported by scientific evidence (https://aopwiki.org/handbooks/5). Therefore, in this study, AOP-Wiki was relied upon to obtain the latest available AOPs.

First, the XML file (released April 1, 2025) was downloaded from the 'Project Downloads' page in AOP-Wiki to obtain the latest information on AOPs. Then, the data was parsed to extract information associated with AOPs like AOP identifier, AOP title, associated KEs (including MIEs and AOs) and KERs, linked stressors, handbook version followed for development of AOP, status according to OECD, and biological applicability information

such as taxonomy, sex and life-stage of the organism, and their corresponding weight of evidence, using an in-house Python script. Additionally, for each KE, information like KE title, KE identifier, level of biological organization, action name, object name, object identifiers, and process name, and information associated with KERs like upstream/downstream KEs, evidence for biological plausibility of KER, adjacency, and the extent of quantitative understanding of KER, were extracted.

5. Curating complete and connected AOPs within AOP-Wiki

AOPs are considered living documents as they are collaboratively developed globally, and are continuously updated based on new scientific evidence (https://aopwiki.org/handbooks/5). Consequently, many AOPs may remain incomplete or lack sufficient information for further analysis. Therefore, based on our previous studies (Sahoo et al., 2024b, 2024a, 2024c), each AOP was systematically checked to filter high quality and complete AOPs. First, AOPs were manually checked and those that comprised KEs with title as 'unknown' or lacked any KEs or KERs were removed. Next, the NetworkX library (Hagberg et al., 2008) was employed in Python to check for disconnected AOPs, and those that contained disconnected components were manually inspected and updated before filtration of AOPs with disconnected components. Finally, the presence of MIE, AO, and a directed path between them was checked, and those that lacked any were filtered out. Through this combined manual and computational effort, 385 complete, connected and high quality AOPs were obtained from AOP-Wiki, which were then designated as 'curated AOPs' (Table S4). These 385 high confidence AOPs comprised 1228 unique KEs (Table S5) and 1966 unique KERs (Table S6).

Subsequently, the AOs of the 385 curated AOPs were classified using the standardized disease ontology from the Disease Ontology database (https://disease-ontology.org/do) (Schriml et al., 2022). This classification resulted in 28 disease classes across 385 AOPs (Table S4). Note that the AOs of 148 out of 385 curated AOPs could not be mapped to any disease ontology term and were therefore designated as 'Unclassified'. Among the 28 disease classes, 'Cancer' emerged as the most represented category, encompassing 46 of the 385 curated AOPs (Table S4).

6. Computation of cumulative weight of evidence (WoE) of AOPs

AOP-Wiki provides a weight of evidence (WoE) information for each KER in an AOP based on its biological plausibility (Table S6). The WoE is a qualitative score

represented in terms of 'High', 'Moderate', 'Low', and 'Not Specified'. Following Ravichandran *et al.* (Ravichandran et al., 2022), the fraction of KERs within an AOP with 'High' WoE [represented as F(High)], 'Moderate' WoE [represented as F(Moderate)], 'Low' WoE [represented as F(Low)] and 'Not Specified' WoE [represented as F(Not Specified)], were computed, and subsequently, a cumulative WoE was assigned to the AOPs based on the following criteria:

i. If F(High) ≥ 0.5, the cumulative WoE of AOP is 'High'
ii. If F(High) < 0.5, but (F(High) + F(Moderate)) ≥ 0.5, the cumulative WoE of AOP is 'Moderate'
iii. If (F(High) + F(Moderate)) < 0.5, but (F(High) + F(Moderate) + F(Low)) ≥ 0.5, the cumulative WoE of AOP is 'Low'
iv. If none of the above criteria are satisfied, the cumulative WoE of AOP is 'Not Specified'

Table S4 provides the cumulative WoE for the curated AOPs.

7. Identification of KEs associated with EDCs

Based on our previous studies (Sahoo et al., 2024b, 2024a, 2024c), different toxicogenomics and biological endpoint information related to EDCs was obtained from AOP-Wiki (https://aopwiki.org), Comparative Toxicogenomics Database (CTD) (Davis et al., 2025) (https://ctdbase.org), DEDuCT (Karthikeyan et al., 2021, 2019) (https://cb.imsc.res.in/deduct/), NeurotoxKb (Ravichandran et al., 2021) (https://cb.imsc.res.in/neurotoxkb/) and ToxCast invitrodb v4.3 (Dix et al., 2007). Based on our previous works (Sahoo et al., 2024b, 2024a), obtained data was then systematically integrated to identify KEs within AOP-Wiki that are linked to EDCs.

7.1. Identification of KEs using AOP-Wiki

AOP-Wiki catalogs information on prototypical stressors for each AOP, based on well-documented associations with these stressors (https://aopwiki.org/handbooks/5). In this study, the information on prototypical stressors associated with AOPs was extracted from the flat download file using an in-house Python script. Thereafter, 100 EDCs were identified as prototypical stressors for 93 AOPs. Subsequently, the 361 KEs within these AOPs were considered as associated with these 100 EDCs (Table S7).

7.2. Identification of KEs using CTD

The Comparative Toxicogenomics Database (CTD) is a comprehensive public resource that links environmental chemicals (C), genes (G), phenotypes (P), and diseases (D) to advance understanding of their effects on health (Davis et al., 2025). This resource facilitates the construction of CGPD-tetramers, which help identify confident associations between chemicals, phenotypes, and diseases, enabling their mapping to KEs within AOP-Wiki. Based on our previous work (Sahoo et al., 2024b, 2024a, 2024c), high confidence CGPD-tetramers associated with EDCs were retrieved and subsequently leveraged to identify KEs within AOP-Wiki.

CTD provides a chemical dictionary linking standard PubChem identifiers and CASRN to chemical IDs within CTD. Initially, a list of 804 unique CTD chemical identifiers was obtained based on overlaps with data compiled in DEDuCTv3.0. Upon closer inspection, it was observed that some of the associations provided by CTD are incorrect. For instance, the EDC Benzo(a)pyrene (DED000576) is linked to two chemical IDs within CTD based on its PubChem identifier (CID:2336). Upon closer inspection, it was observed that one of the chemical IDs, MESH:D001580 corresponds to the class of benzopyrenes, while MESH:D001564 corresponds to Benzo(a)pyrene. Therefore, all the associations were manually screened to remove non-specific information within CTD, resulting in the identification of 783 chemical IDs within CTD associated with 764 EDCs within DEDuCTv3.0.

Next, the CTD November 2025 release was accessed and the CGPD-tetramers were constructed, wherein the following were considered: (i) chemical-gene and chemical-phenotype associations that are relevant to humans and rodents, and are supported with literature evidence; (ii) chemical-disease and gene-disease associations with 'marker/mechanism' evidence, and are supported with literature evidence; (iii) gene-phenotype associations with GO annotations based on only the experimental results in human or rodent-based studies (https://geneontology.org/docs/guide-go-evidence-codes/). This process resulted in a list of 208809 CGPD-tetramers comprising 364 EDCs, 3311 genes, 1747 phenotypes, and 858 diseases.

Thereafter, phenotypes and diseases were mapped to the KEs within AOP-Wiki following the procedure described in our previous works (Sahoo et al., 2024b, 2024a). First, the CGPD-tetramer phenotype GO terms directly associated with a KE through the process

identifier provided by AOP-Wiki were considered as directly mapped to the corresponding KE. GO organizes terms in a directed acyclic graph structure, where parent and child node relationships can be used to infer the similarity of biological events (Ashburner et al., 2000). Therefore, the immediate neighboring GO terms for the CGPD-tetramer phenotype GO terms were generated using the GOSim package (Fröhlich et al., 2007) in the R programming language. These neighboring GO terms were then overlapped with the KE-associated process identifier to identify preliminary associations. Subsequently, the KE titles and phenotype names were manually screened to retain associations that matched the biological process outlined in the KE title and KE information page. As an example, KE 'Increased, locomotion' (KE:622), and 'Decreased, locomotion' (KE:635) were both assigned the process 'locomotion' (GO:0040011) in the KE information file obtained from AOP-Wiki. From the CTD chemical-phenotype associations, phenotypes such as 'positive regulation of locomotion' (GO: GO:0040017) and 'negative regulation of locomotion' (GO:0040013), were identified as chemical-induced phenotypes from rodent-based studies. Upon further inspection, these phenotypes were identified as child nodes of the GO 'locomotion'. Therefore, these phenotypes were first mapped to both the KEs based on the GO 'locomotion'. Thereafter, upon inspecting the associations, we clearly note that 'Increased, locomotion' maps only to 'positive regulation of locomotion' while 'Decreased, locomotion' maps only to 'negative regulation of locomotion', and therefore we filtered out the other associations. Through this process, 310 KEs were linked to 279 phenotypes across 347 EDCs (Table S7).

For disease mapping, the disease-associated MeSH terms were first obtained from the CGPD-tetramer disease terms. For each term, the corresponding title and metadata information were screened using the MeSH browser (https://meshb.nlm.nih.gov/) and associated with the relevant KE based on underlying biological processes. For example, 'Parkinson Disease' (MESH:D010300) was mapped to KEs such as 'Neuronal dysfunction' (KE:191), 'Altered, Neurophysiology' (KE:192), and 'Neurological disorder' (KE:2150), as these KEs capture the loss of neurons and the resulting neurological deficits identified from the disease descriptors and scope notes in the MeSH browser. Through this process, 256 KEs were linked to 548 diseases across 358 EDCs (Table S7).

7.3. Identification of KEs using DEDuCTv3.0 endocrine-mediated endpoints and NeurotoxKb neurotoxicant endpoints

Based on previous studies (Sahoo et al., 2024b, 2024a, 2024c), the 796 endocrine-mediated endpoints associated with the 1043 potential EDCs within DEDuCTv3.0 were first screened to identify relevant KEs within AOP-Wiki. In particular, the endpoints and the KE titles were manually inspected, and mapped based on biological similarity. For example, endocrine-mediated endpoints such as 'Induce ferroptosis in ovary', 'Induce ferroptosis in pancreas', 'Induce ferroptosis in testis', and 'Induce ferroptosis in liver' were mapped to KEs such as 'Increase, Ferroptosis' (KE:1994) and 'Ferroptosis related pathways activation' (KE:2241), as these KEs capture the underlying biological event shared across the endpoints. Similarly, the endpoint 'Increased adiposity' was mapped to KEs such as 'Increased adipocyte size' (KE:2127), 'Increased adipocyte numbers' (KE:2128), 'Increased, adipogenesis' (KE:1029), and 'Increased adipogenesis' (KE:1449), as they collectively reflect the biological processes associated with adiposity. This process resulted in the identification of 276 KEs associated with 361 endocrine-mediated endpoints, across 927 EDCs (Table S7).

NeurotoxKb (Ravichandran et al., 2021) (https://cb.imsc.res.in/neurotoxkb/) is a manually curated resource focusing on mammalian neurotoxicity endpoints associated with environmental chemicals, compiled from published literature. In this study, the neurotoxic endpoints corresponding to EDCs within NeurotoxKb were extracted and subsequently considered to identify relevant KEs within AOP-Wiki. The endpoints catalogued in NeurotoxKb were already represented as MeSH-based terms. Therefore, the mapping criteria applied were analogous to those followed for mapping CTD disease terms to KEs. Briefly, the neurotoxic endpoints and KE titles in AOP-Wiki were manually inspected to identify 64 KEs associated with 55 neurotoxic endpoints across 123 EDCs (Table S7).

7.4. Identification of KEs using ToxCast

ToxCast is a program by the United States Environmental Protection Agency (US EPA) designed to enhance chemical toxicity predictions through *in vitro* high-throughput screening of various environmental chemicals (Dix et al., 2007). These *in vitro* approaches provide crucial information on the associated biological processes and gene alterations triggered by stressor interactions, which can aid in the identification of potential MIEs related to active chemicals (Sahoo et al., 2024b, 2024a, 2024c; Knapen et al., 2018; Aguayo-Orozco et al., 2019; Jeong and Choi, 2019). In this study, ToxCast invitrodb v4.3 (Feshuk et al., 2023; US EPA, 2025) was utilized to identify KEs associated with EDCs.

First, the chemicals and their corresponding assay information were extracted from the 'mc5-6_winning_model_fits-flags_invitrodbv4_3_AUG2024.csv' file, and active assay endpoints for each chemical (defined as 'hitc' ≥ 0.9) were identified (Feshuk et al., 2023). Next, the 'top' value of the corresponding winning model from the 'mc4_all_model_fits_invitrodbv4_3_AUG2024.Rdata' file was utilized to determine whether these active chemicals exhibited an 'activatory' or 'inhibitory' effect (Feshuk et al., 2023).

To ensure that these active endpoints are not due to non-specific activation of reported genes (referred to as 'cytotoxicity-associated burst' phenomenon),(Judson et al., 2016) the following Z-score statistic proposed by Judson *et al.* (Judson et al., 2016) was applied:

$$Z(chemical, assay) = \frac{-logAC_{50}(chemical,assay) - median[-logAC_{50}(chemical,cytotox)]}{\text{global cytotoxicity MAD}}$$

Here, 'logAC$_{50}$(chemical,assay)' is the logarithm of the AC$_{50}$ value of the chemical in the assay, 'logAC$_{50}$(chemical,cytotox)' is the logarithm of the AC$_{50}$ value of the chemical in the corresponding cytotoxicity assay, and the 'global cytotoxicity MAD' is the median of the median average deviations (MAD) of the logAC$_{50}$(chemical,cytotox) distributions across all chemicals. Based on this definition, Z-scores ranging between +3 and -3 are considered indicative of cytotoxicity-associated bursts (Judson et al., 2016).

In this study, the global cytotoxicity MAD and logAC$_{50}$(chemical,cytotox), given by the column titled 'cytotox_median_log', were retrieved from the 'cytotox_invitrodb_v4_3_AUG2024.xlsx' file to identify the cytotoxicity-associated bursts associated with EDCs (Feshuk et al., 2023). The assays with Z-scores between +3 and -3 were discarded, and the resulting assays were subsequently filtered to obtain human- or rodent-specific endpoints, and then mapped to KEs within AOP-Wiki. Thereafter, the gene targets associated with each ToxCast assay were obtained from the 'assay_annotations_invitrodb_v4_3_AUG2024.xlsx' file, which formally documents the intended gene target for each assay endpoint.

The ToxCast assay endpoints and the KE titles from the AOP-Wiki were manually inspected to identify relevant KEs. For example, the assay 'ATG_ERE_CIS' (aeid:75) showed an 'inhibitory' response across chemicals within DEDuCTv3.0. This assay endpoint is linked to the estrogen receptor ESR1, where an inhibitory signal means that the gene's function is inhibited, i.e., the chemicals act as antagonists on ESR1. Therefore, this assay endpoint and outcome were mapped to 'Antagonism, Estrogen receptor' (KE:112), 'Suppression, Estrogen receptor (ER) activity' (KE:1046), and 'Estrogen receptor alpha

inactivation' (KE:2126). This manual curation led to the identification of 160 KEs associated with 318 assay endpoints across 523 EDCs (Table S7).

Altogether, 955 KEs were identified to be associated with 984 EDCs through the integration of heterogeneous information on toxicogenomic and biological endpoints from five exposome-relevant resources, namely, AOP-Wiki, CTD, DEDuCT, NeurotoxKb, and ToxCast (Table S7).

8. Construction of stressor-AOP network

Stressor-AOP network provides a broader perspective on the impacts of stressors across diverse biological processes by linking the stressors to AOPs (Aguayo-Orozco et al., 2019; Sahoo et al., 2024a). To better understand the perturbances caused by EDCs, a stressor-AOP network was constructed, linking EDCs to different AOPs within AOP-Wiki. In order to obtain high confidence associations between the chemicals and AOPs, only the 385 curated AOPs were relied upon (Table S4).

First, the EDCs were linked to any AOP if they shared at least one common KE. Thereafter, these links were characterized based on two criteria namely, the coverage score and the level of relevance. The coverage score of a stressor-AOP link is defined as the ratio of the number of KEs within that AOP associated with the stressor to the total number of KEs within that AOP (Chai et al., 2021; Sahoo et al., 2024b). This score is a real-valued number between 0 and 1, and is denoted as the edge weight of the linkage between a stressor and an AOP in the constructed stressor-AOP network. The level of relevance is a qualitative score used to identify the relevance of stressor-AOP association within the stressor-AOP network (Sahoo et al., 2024a), and this score is denoted as an attribute of the edge in the constructed stressor-AOP network. Level of relevance is a five-level criterion defined as follows:

- *Level 1*: The stressor is associated with at least one KE within an AOP, where the KE is neither MIE nor AO within that AOP
- *Level 2*: The stressor is associated with at least one AO within an AOP, but not associated with any MIE within that AOP
- *Level 3*: The stressor is associated with at least one MIE within an AOP, but not associated with any AO within that AOP
- *Level 4*: The stressor is associated with at least one MIE and one AO within an AOP
- *Level 5*: The stressor is associated with at least one MIE and one AO within an AOP, and there exists a directed path between the associated MIE and AO

Table S8 contains all the data on the stressor-AOP network constructed for EDCs in DEDuCTv3.0, including the coverage score and level of relevance for each of the stressor-AOP links.

9. Construction of a toxicology knowledge graph for EDCs in DEDuCTv3.0

In this study, a toxicology knowledge graph was constructed by linking the chemicals (EDCs) and endocrine-mediated endpoints within DEDuCTv3.0 with gene, phenotype, disease, biological key events (KEs), and adverse outcome pathways (AOPs) from toxicology-relevant resources such as ToxCast, CTD, and AOP-Wiki (Figure 2). We refer to this knowledge graph for EDCs as 'DEDuCT-KG', and the following subsections summarize the steps involved in curation of links and nodes within the knowledge graph.

9.1. Curation of links between chemicals and endocrine-mediated endpoints

The endocrine-mediated endpoints associated with chemicals have been systematically curated from published literature in DEDuCTv3.0, and therefore can be considered as high confidence links. All these chemical-endpoint links have been assigned the edge type 'reported_to_cause' to better explain these links in the knowledge graph. Through this process, 10532 links were identified that connected 1043 EDCs with 796 endocrine-mediated endpoints in DEDuCT-KG.

9.2. Curation of links between chemicals and genes from CTD and ToxCast

CTD provides curated chemical-gene interactions obtained from published literature, where it compiles additional information on the gene form of the interacting gene, corresponding organism name, taxonomic identifier, chemical-gene interaction actions, and associated PMID of the published literature. First, the NCBI taxonomy browser (https://www.ncbi.nlm.nih.gov/taxonomy) was searched with the corresponding organism identifiers to obtain information on the taxonomic classification such as the domain, clade, kingdom, phylum, and order, among others. To obtain human- or rodent-relevant gene information, only species such as *Homo sapiens* (organism id: 9606) and those belonging to the order 'Rodentia' were selected. This resulted in the identification of 28 relevant species. Next, genes were filtered based on their gene form information to obtain those entries that were annotated as gene, mRNA, or protein. Further, genes were filtered out based on the associated interaction actions such as 'abundance', 'synthesis', 'response to substance', 'reaction', 'cotreatment', 'transport', 'localization', 'degradation', 'export', 'import',

‘metabolic processing’, ‘mutagenesis’, ‘secretion’, ‘uptake’, ‘stability’, and only those with terms such as ‘expression’ and ‘activity’ were retained. These interaction actions further included the kind of actions, such as ‘increases’, ‘decreases’, or ‘affects’. For some of the chemical-gene links, it was observed that there were different curated interaction actions across different studies. Therefore, to mitigate such discrepancies, the following standardization criteria was followed: (i) if the interaction action included terms such as both ‘increases’/’decreases’ and ‘affects’, then it was assigned to ‘increases’/’decreases’ respectively; and (ii) in case both ‘increases’ and ‘decreases’ were present, then it was assigned as ‘affects’. Through this systematic standardization process, 509996 links were identified that connected 682 chemicals to 29560 genes. The standardized interaction actions were assigned as the edge types within the knowledge graph.

Chemical-gene interactions were also obtained from human- or rodent-relevant data curated from ToxCast invitrodb v4.3. Here, the endpoints where the response was characterized as either ‘activatory’ or ‘gain’ were assigned as ‘reported_to_activate’, and those characterized as ‘inhibitory’ or ‘loss’ were assigned as ‘reported_to_inhibit’ to better explain the chemical-target gene associations from ToxCast *in vitro* assays. Through this process, 13205 links were identified that connected 579 chemicals to 400 genes. The standardized response information was assigned as edge types within the knowledge graph.

9.3. Curation of links between chemicals and phenotypes or diseases from CTD

CTD provides curated chemical-phenotype interactions obtained from published literature, where it compiles additional information on the corresponding organism name and taxonomic identifier, chemical-phenotype interaction actions, and associated PMID of the published literature. Similar to chemical-gene links, these interactions were first filtered to obtain human- or rodent-relevant information. Subsequently, the discrepancies in chemical-phenotype interactions were resolved, resulting in the identification of 28561 links that connected 599 chemicals to 4431 phenotypes. The standardized interaction actions were assigned as the edge types within the knowledge graph.

CTD also provides curated chemical-disease associations, where it additionally annotates the association as either ‘marker/mechanism’ or ‘therapeutic’ based on the evidence from corresponding published literature. According to the CTD terminology, ‘marker/mechanism’ corresponds to evidence suggesting that the chemical exposure correlates with the disease or may play a role in the etiology of the disease, while

'therapeutic' corresponds to evidence suggesting the therapeutic potential of the chemical in a disease (https://ctdbase.org/help/glossary.jsp). Therefore, only associations marked as 'marker/mechanism' were obtained, resulting in the identification of 16219 links connecting 586 chemicals to 1805 diseases. The edge type was assigned as 'causes_or_correlated_with' to better explain the chemical-disease associations in the knowledge graph.

9.4. Curation of links between genes and disease within CTD

CTD provides curated gene-disease associations, where it additionally annotates the association as either 'marker/mechanism' or 'therapeutic' based on the evidence from corresponding published literature. According to CTD terminology, 'marker/mechanism' corresponds to evidence suggesting that the gene may be a biomarker for the disease or may play a role in the etiology of the disease, while 'therapeutic' corresponds to evidence suggesting the gene may be a potential therapeutic target in the treatment of the disease (https://ctdbase.org/help/glossary.jsp). Therefore, only associations marked as 'marker/mechanism' were obtained, resulting in the identification of 27598 links connecting 7772 genes to 2466 diseases. The edge type was assigned as 'associated_with' to better explain the gene-disease associations in the knowledge graph.

9.5. Curation of links between genes and phenotypes from Gene Ontology

Gene Ontology (GO) provides annotated gene-phenotype associations, including the taxonomy identifier of the organism in which the association was identified, the kind of evidence, and a qualifier term to explain the association between the gene and phenotype. First, the GO annotations were obtained from the NCBI gene resource (https://ftp.ncbi.nih.gov/gene) (last accessed on November 26, 2025), and only gene-phenotype associations with experimental results were retained (https://geneontology.org/docs/guide-go-evidence-codes/). Next, the links were filtered to retain only human- or rodent-relevant associations based on taxonomic information. It was observed that each link was annotated with a qualifier term that defines the gene-phenotype association. The links were then filtered to remove qualifier terms that included 'NOT', resulting in the identification of 304398 links connecting 36463 genes to 18311 phenotypes. The qualifier terms were then assigned as the edge types within the knowledge graph.

9.6. Curation of links between endocrine-mediated endpoints and phenotypes or diseases

In DEDuCTv3.0, endocrine-mediated endpoints were manually curated based on the biological endpoints reported in the supporting information. Mapping these endpoints to phenotypes and disease terms enables making the data more FAIR-compliant (Wilkinson et al., 2016). For this purpose, the definitions of the endpoints, phenotypes, and disease terms were manually checked and mapped based on underlying biological similarities. Through this process, 581 links were identified that connected 394 endocrine-mediated endpoints to 248 phenotypes and 78 diseases. These links have been assigned the edge type 'related_to' to better explain these links in the knowledge graph.

9.7. Curation of links between chemicals, genes, phenotypes, diseases, endocrine-mediated endpoints, and key event information within AOP-Wiki

Integrating diverse toxicological data with information contained within the AOP-Wiki enables the exploration of diverse toxicity mechanisms associated with chemicals (Chivukula et al., 2025; Sahoo et al., 2024b, 2024a, 2024c). Furthermore, integrating these data within a knowledge graph enables the exploration of previously missed associations and may potentially aid in elucidating the chemical-associated toxicity mechanisms that were otherwise less understood. Based on the procedure explained in the section 7 of the Supplementary Information titled 'Identification of KEs associated with EDCs', all the endocrine-mediated endpoints, phenotypes, and diseases were systematically mapped to KE information within AOP-Wiki. This resulted in the identification of 2334 links connecting 411 phenotypes, 733 diseases, 320 endpoints, with 632 KEs within AOP-Wiki. These links have been assigned the edge type 'related_to' to better explain these links in the knowledge graph. Next, the curated prototypical stressor information from AOP-Wiki was utilized to identify 175 links connecting 100 chemicals to 93 AOPs. These links have been assigned the edge type as 'prototypical_stressor_of' to better explain these links in the knowledge graph.

Further, the gene-KE associations were obtained from data curated by Saarimäki *et al.* (Saarimäki et al., 2023) (https://doi.org/10.5281/zenodo.7980953). In this dataset, the provided Ensembl gene IDs were mapped to Entrez gene identifiers using the BioMart online tool (https://asia.ensembl.org/info/data/biomart/index.html?) (Last accessed on November 26, 2025), resulting in the identification of 148171 links connecting 969 KEs and 14298 genes. These links have been assigned the edge type 'mapped_to' to better explain these links in the knowledge graph.

9.8. Curation of links between KEs and AOPs within AOP-Wiki

The associations between KEs and AOPs was obtained from AOP-Wiki, resulting in 2566 links connecting 1228 KEs linked to 385 AOPs. Table S6 lists these associations and specifies which KEs are MIEs and which are AOs within each AOP. To better explain these links in the knowledge graph, they have been assigned the edge type 'part_of'. Since the designation of a KE as an MIE or AO is specific to a given AOP (https://aopwiki.org/handbooks/5), this information is recorded as an attribute of the corresponding edge in the knowledge graph.

9.9. Curation of links between chemicals based on structural similarity

To enable chemical similarity-based search across DEDuCT-KG, the chemical similarities were computed between EDCs as Tanimoto coefficient based on their ECFP4 fingerprints (Bajusz et al., 2015; Karthikeyan et al., 2019). Subsequently, the chemical pairs having Tanimoto coefficients > 0.5 were considered as having structural similarity. This resulted in the identification of 1609 links connecting 634 chemicals within DEDuCTv3.0. These links have been assigned the edge type 'structurally_similar_to' to better explain these links in the knowledge graph.

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Supplementary Figures

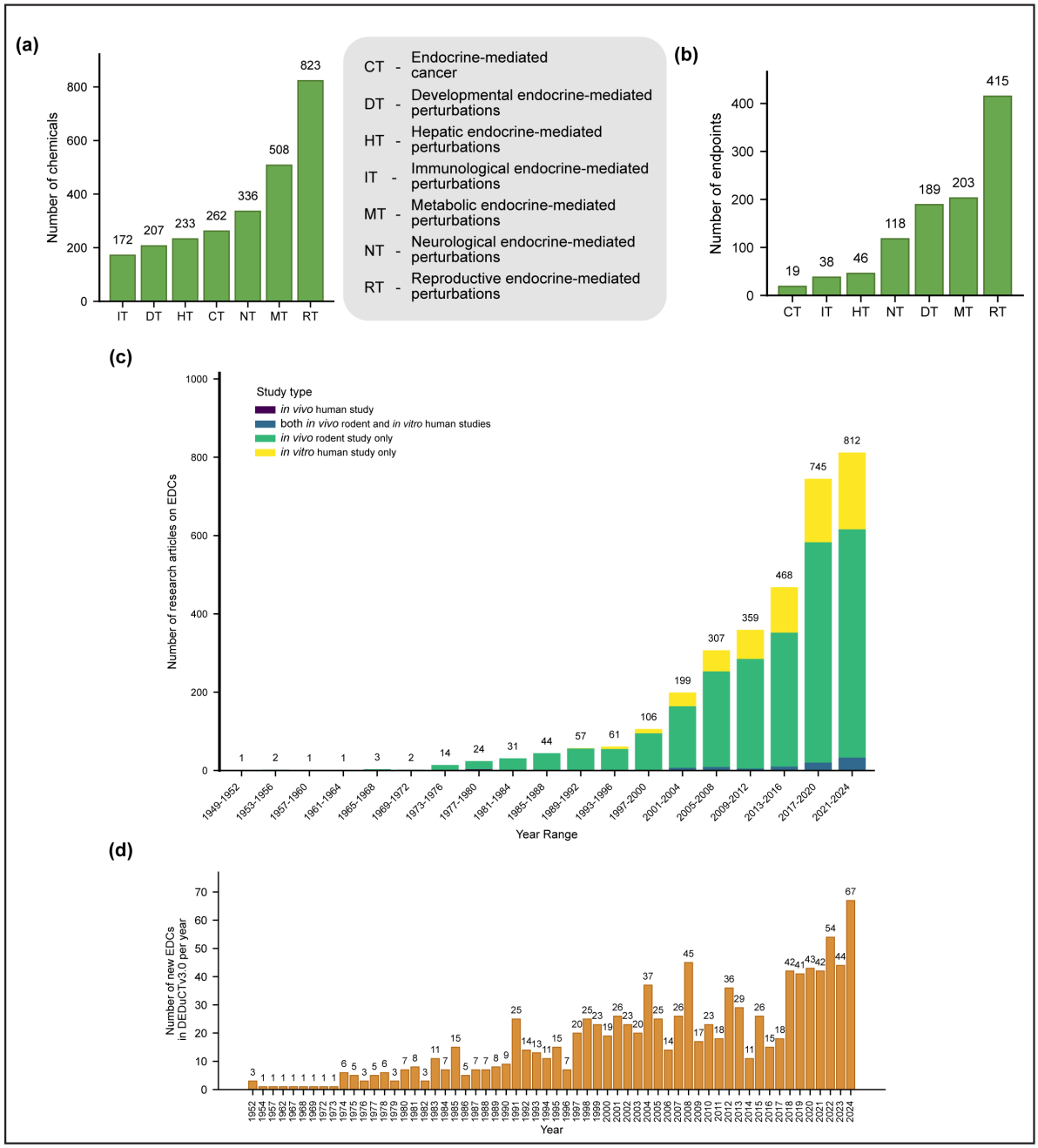


Figure S1. (**a**) The distribution of the 796 endocrine-mediated endpoints in DEDuCTv3.0 into the seven systems-level perturbation categories. (**b**) The distribution of chemicals within DEDuCTv3.0 into the seven systems-level perturbation events based on their associated endocrine-mediated endpoint obtained from published literature. (**c**) A chronological analysis of the 3269 published articles supporting 1043 EDCs in DEDuCTv3.0. Note, 32 articles published in January of 2025 are not included. (**d**) Bar plot depicting number of new EDCs identified per year based on published literature compiled in DEDuCTv3.0. Note, 4 new EDCs from January 2025 are not included.

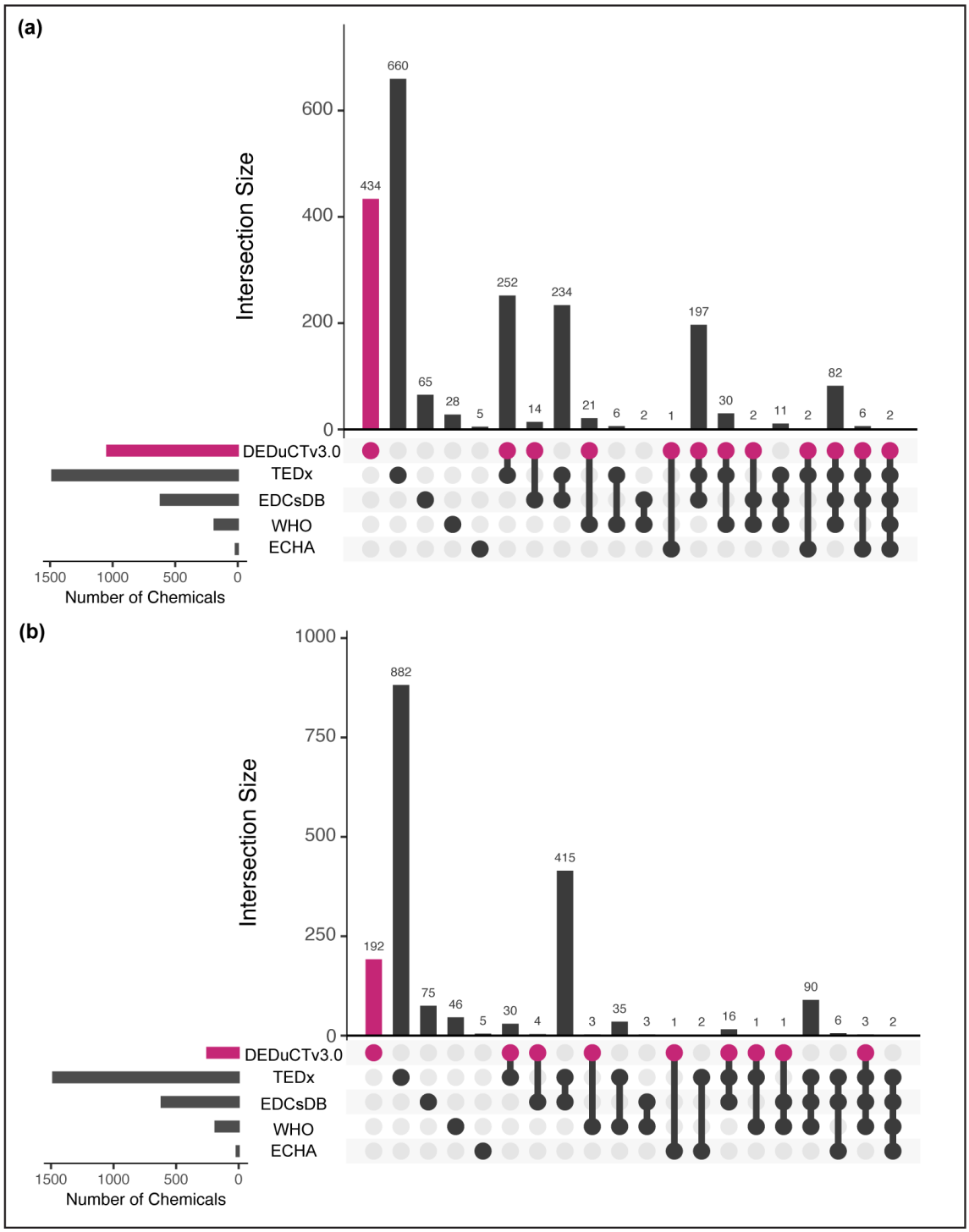


Figure S2. UpSet plot representation of unique overlaps between EDCs in DEDuCTv3.0 and other EDC-relevant databases such as the Endocrine Disruption Exchange (TEDx) (1482 EDCs), EDCs Databank version 2015 (EDCsDB) (615 EDCs), World Health Organization (WHO) report (184 EDCs), European Chemical Agency (ECHA) conclusive human EDCs from EDC assessment list (16 EDCs). (**a**) Comparison with all 1043 EDCs compiled in DEDuCTv3.0. (**b**) Comparison with the 251 new EDCs compiled in DEDuCTv3.0. These UpSet plots were generated using UpSetR package in R.

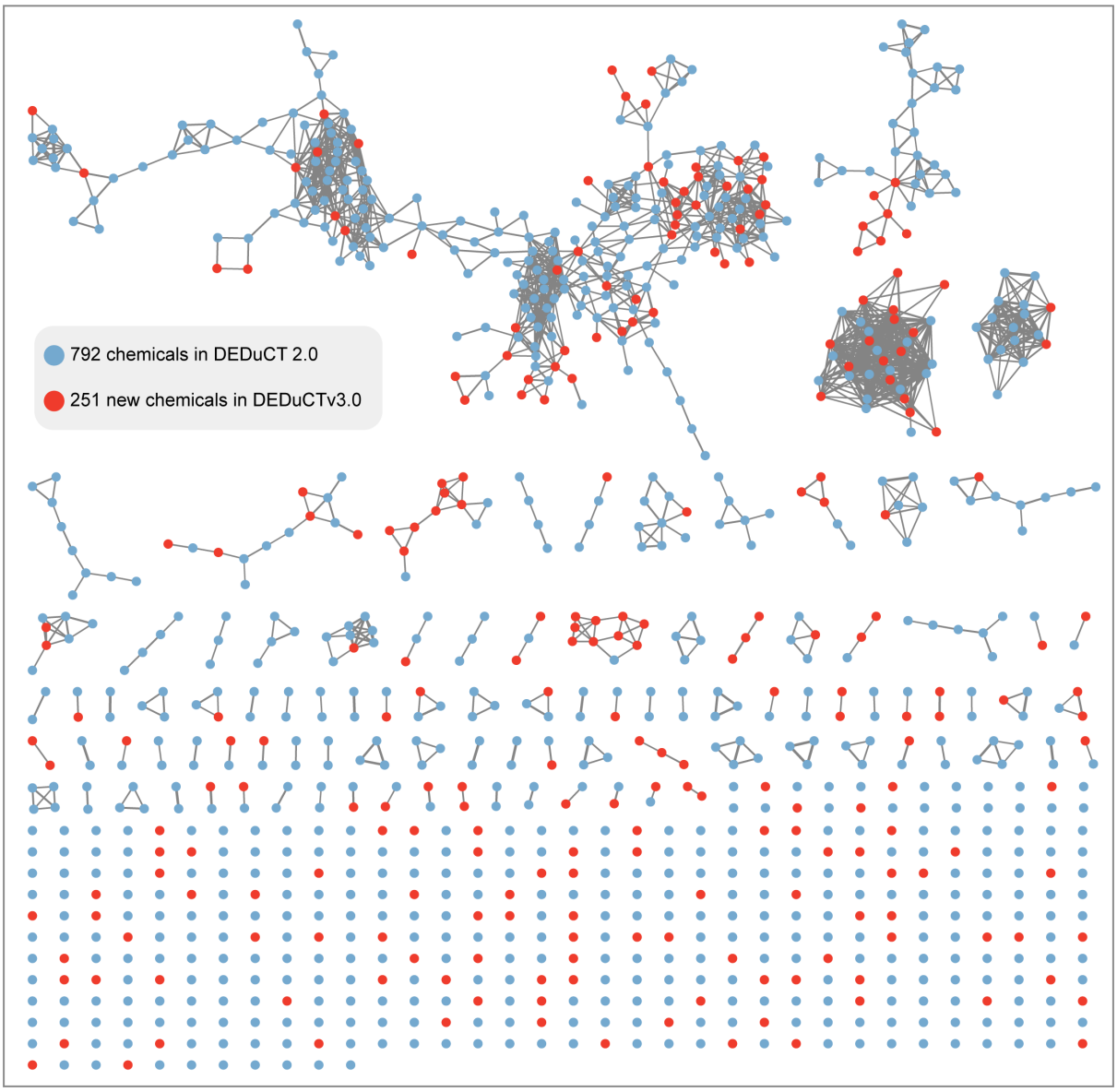


Figure S3. Network visualization of the chemical similarity network (CSN) constructed based on the Tanimoto coefficients computed for each pair of chemicals using ECFP4 fingerprints. In this figure, edges are filtered to represent only those with Tanimoto coefficient > 0.5, with the edge thickness proportionally scaled to the corresponding values. The chemicals compiled in DEDuCT 2.0 are represented in blue, while the 251 new chemicals curated in DEDuCTv3.0 are shown in red.

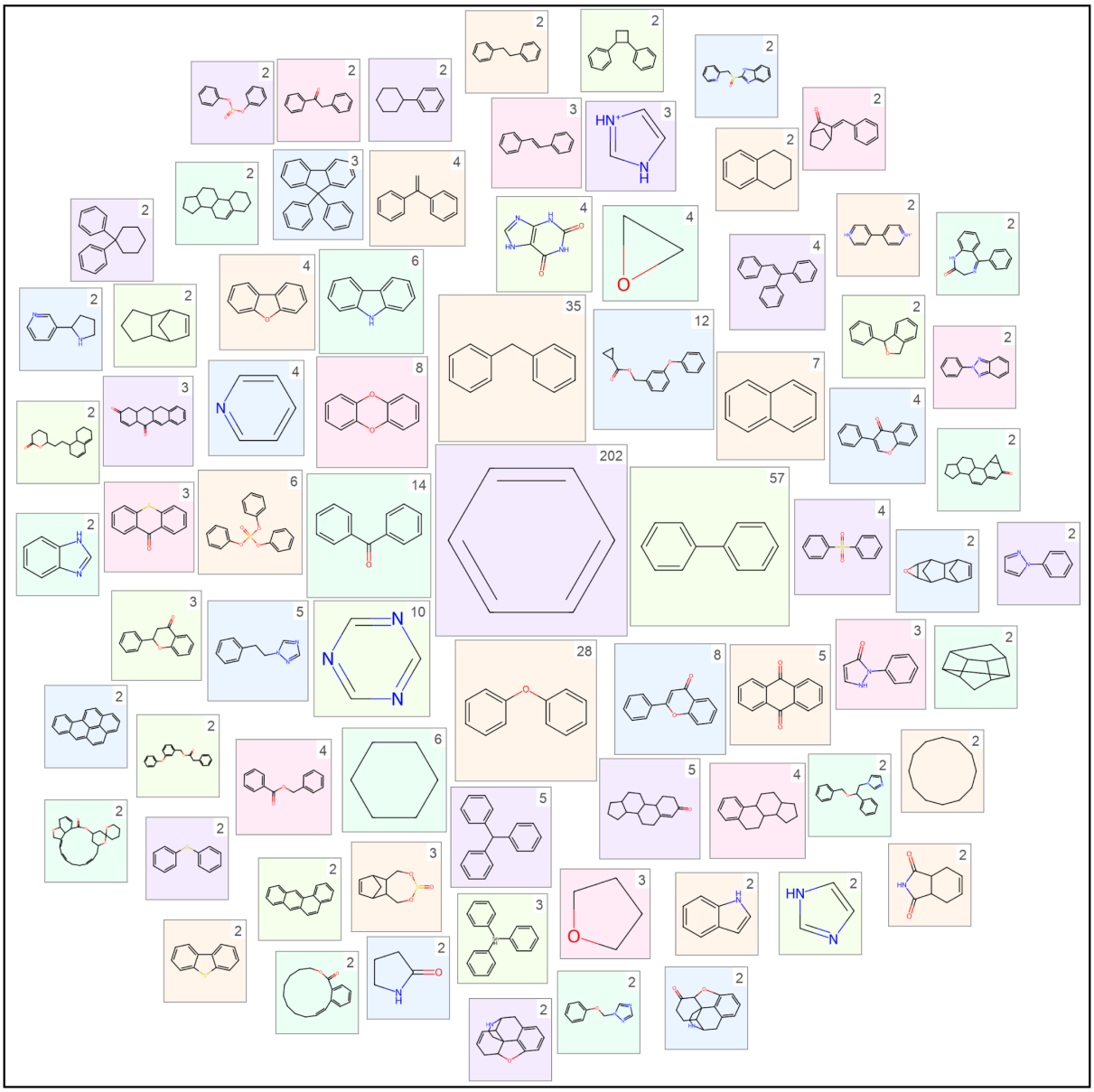


Figure S4. A scaffold cloud representation of the 75 unique scaffolds that are shared between at least two chemicals. The scaffolds are computed using RDKit based on the Bemis-Murcko definition. The number of chemicals sharing the scaffold is mentioned at the top-right corner of each box containing the scaffold.

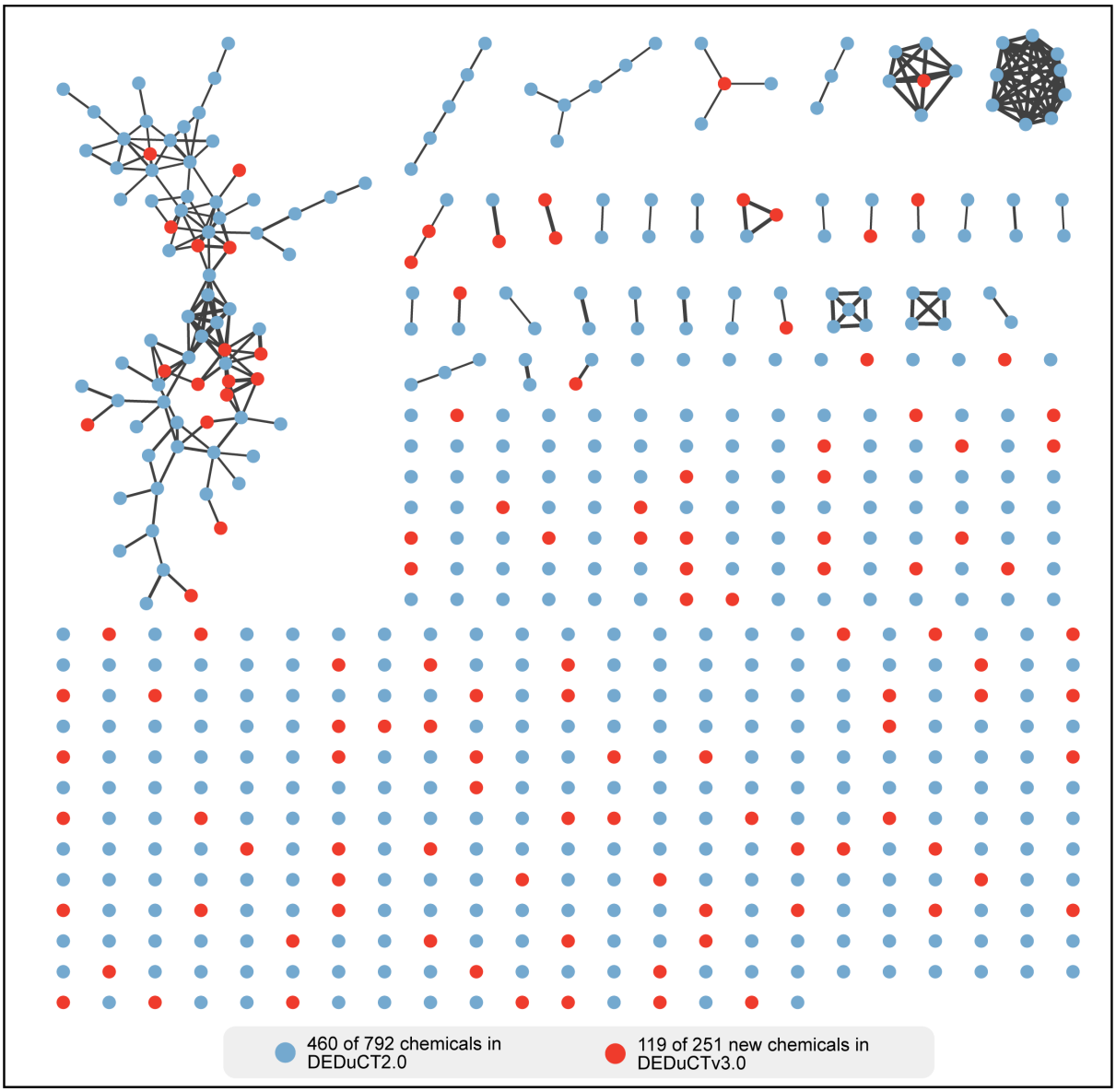


Figure S5. Network visualization of the target similarity network (TSN) constructed based on the Jaccard indices computed for each pair of chemicals using information on gene targets. In this figure, edges are filtered to represent only those with Jaccard index > 0.5, with the edge thickness proportionally scaled to the corresponding values. The 460 of 792 chemicals compiled in DEDuCT 2.0 are represented in blue, while the 119 of 251 new chemicals curated in DEDuCTv3.0 are shown in red.

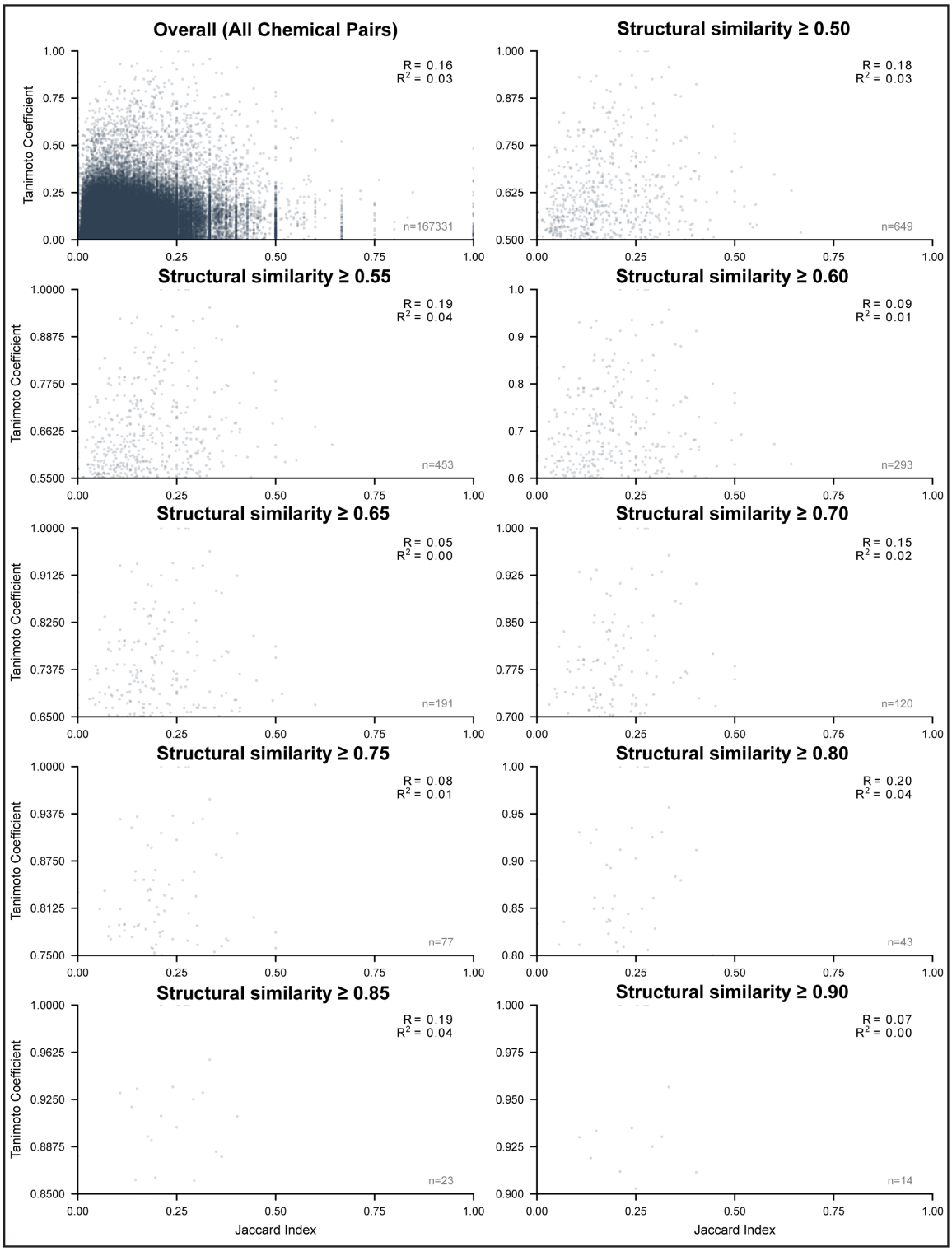


Figure S6. The scatter plot of target similarity vs structure similarity between pairs of EDCs. Here, the structure similarity is given by Tanimoto coefficient, and target similarity is given by Jaccard Index. The plots are further divided based on different structural similarity cut-off values. The Pearson correlation coefficient (R) and the coefficient of determination (R^2) values are denoted at the top-right of each graph. Further, the number of chemical pairs in each graph is denoted at the bottom right.

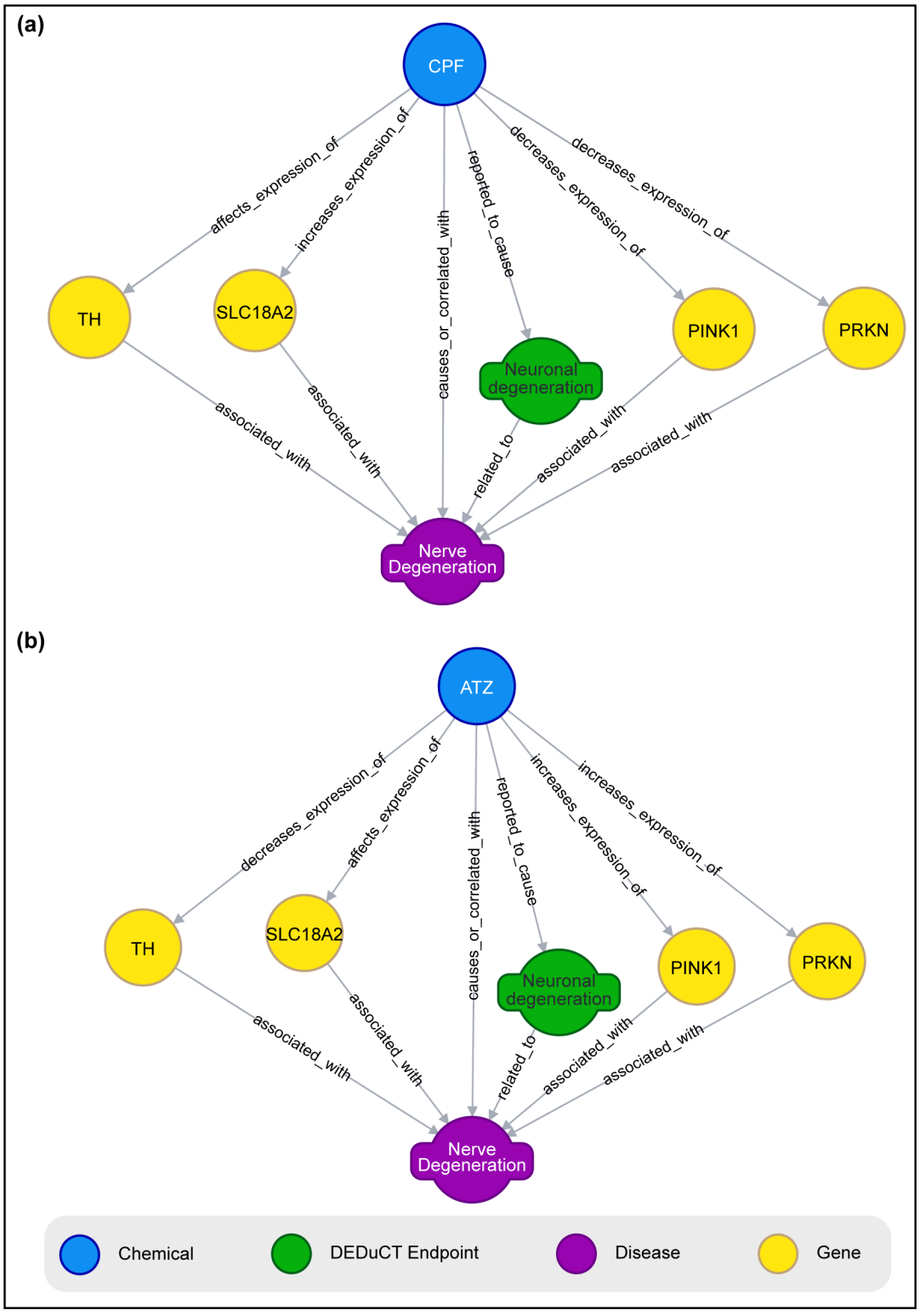


Figure S7. Chemical-gene-disease-endpoint subgraphs depicting the genes, diseases, and endocrine-mediated endpoint obtained from DEDuCT-KG, associated with Parkinson's disease: (a) chlorpyrifos (CPF; CAS:2921-88-2); (b) atrazine (ATZ; CAS:1912-24-9). These subgraphs were generated using Neo4j.

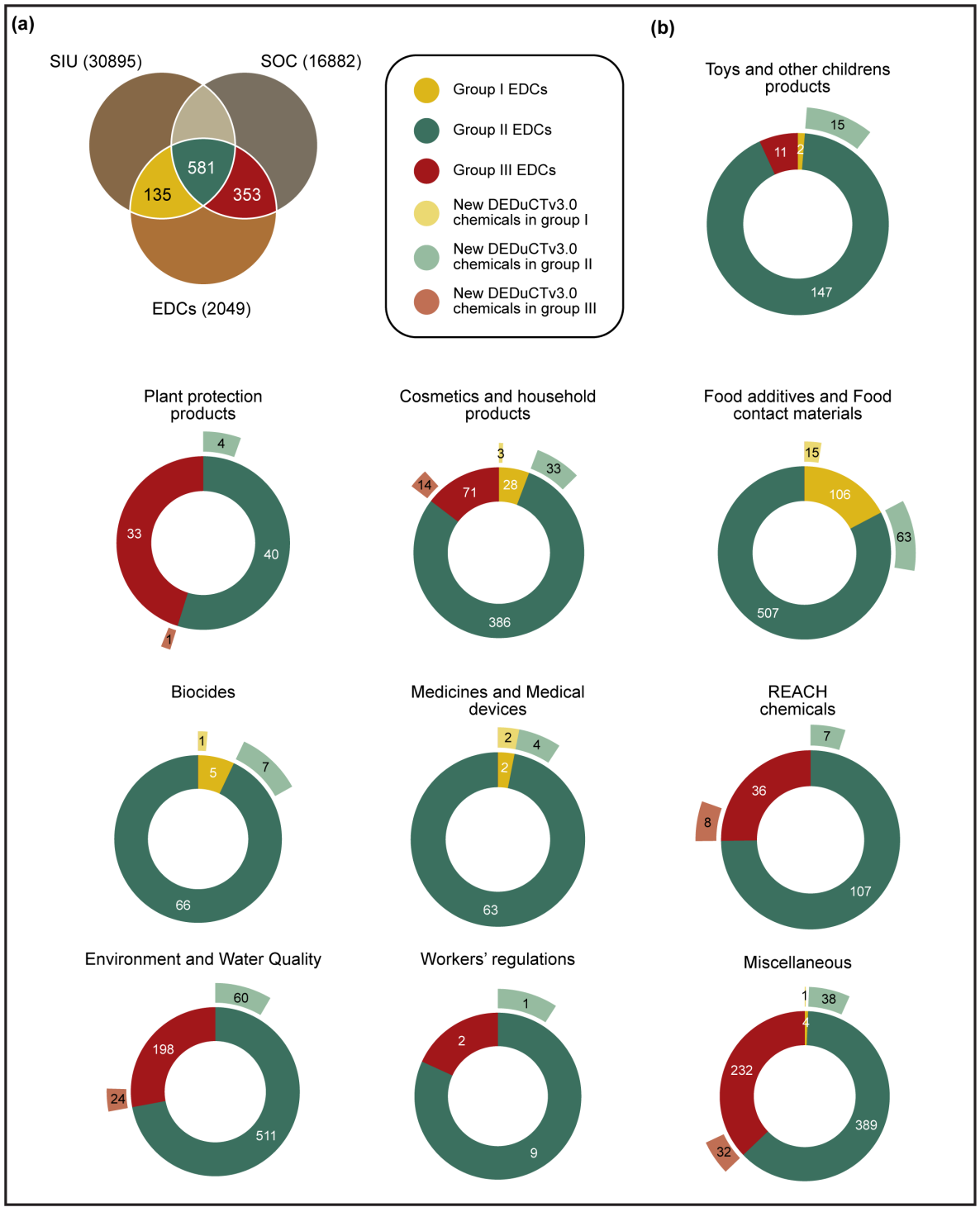


Figure S8. Distribution of 2049 unique potential EDCs from DEDuCTv3.0, TEDx, EDCs Databank, and WHO report across 54 chemical lists. **(a)** Venn diagram depicting chemicals categorized into groups I, II, and III. **(b)** Sunburst plots depicting distribution of potential EDCs across 10 categories of chemical lists. Within each category, the inner ring gives the number of potential EDCs categorized as groups I, II, or III, and the outer ring depicts the number of newly compiled chemicals within DEDuCTv3.0 present in each group.

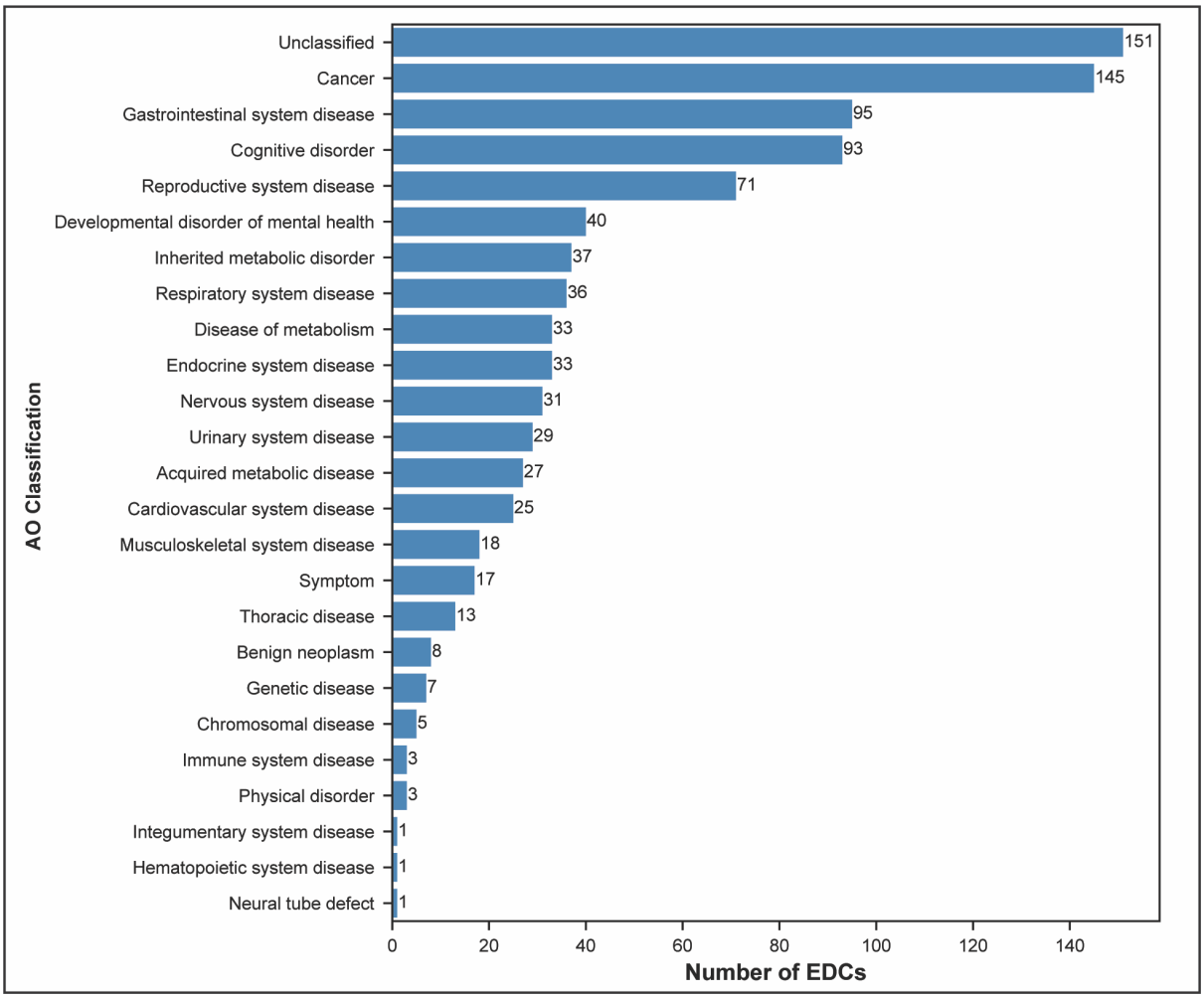


Figure S9. Distribution of 250 EDCs across 28 disease categories based on the AO classification of their associated AOPs at Level 5 relevance.

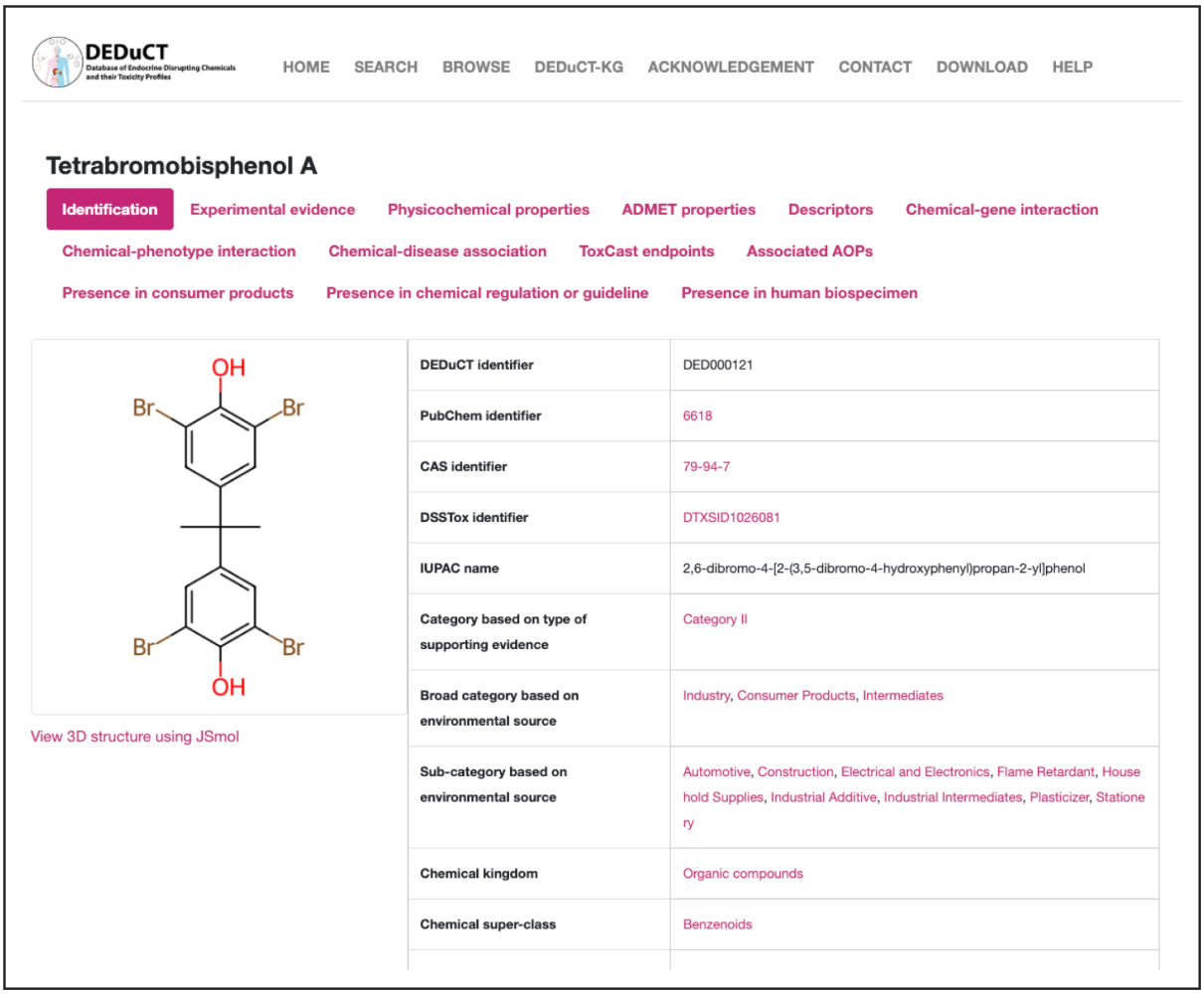


Figure S10. Screenshot for the updated chemical information page in DEDuCTv3.0 webserver.

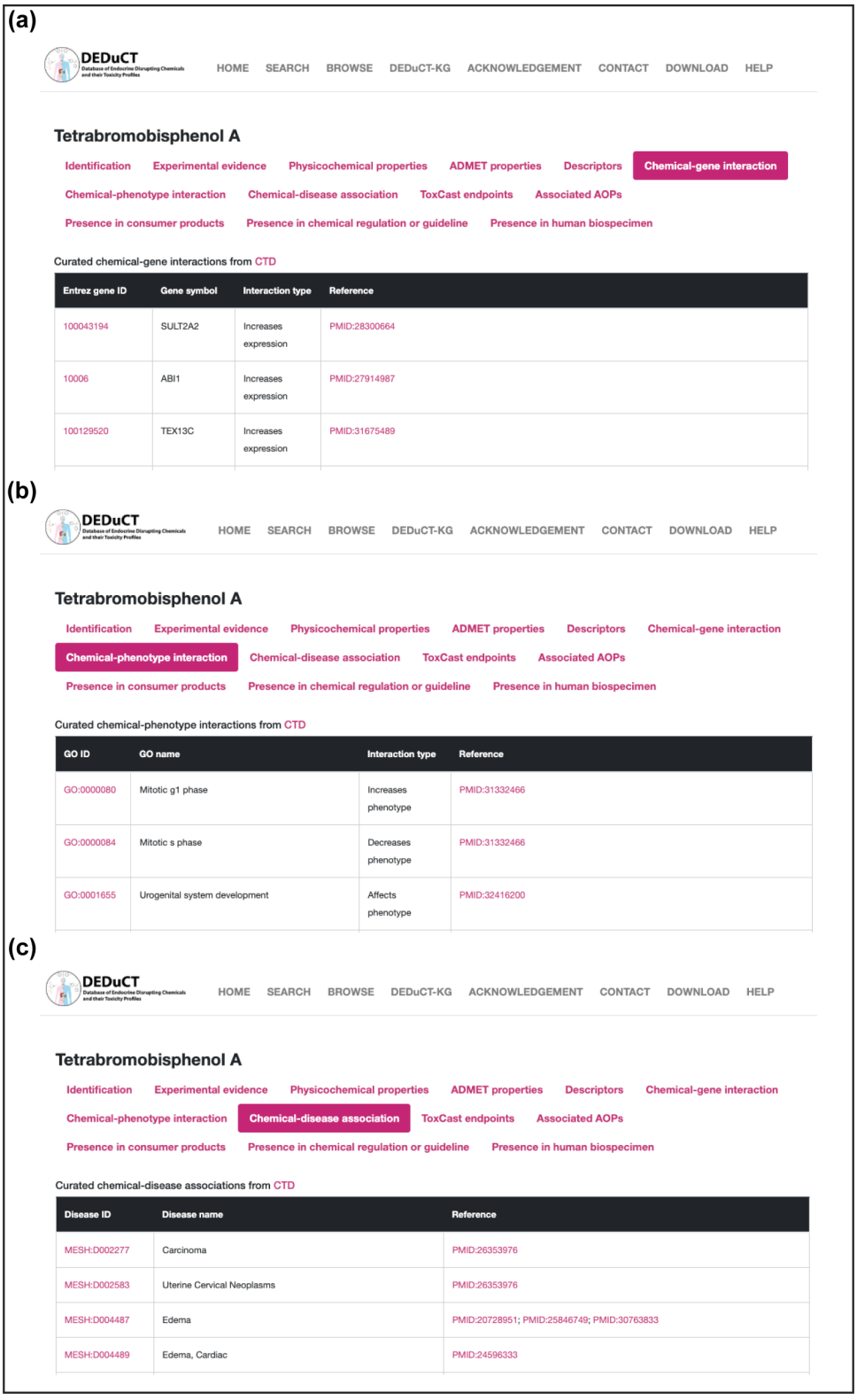


Figure S11. Screenshots from DEDuCTv3.0 webserver of the chemical information pages on: (**a**) chemical-gene interactions; (**b**) chemical-phenotype interactions; (**c**) chemical-disease associations.

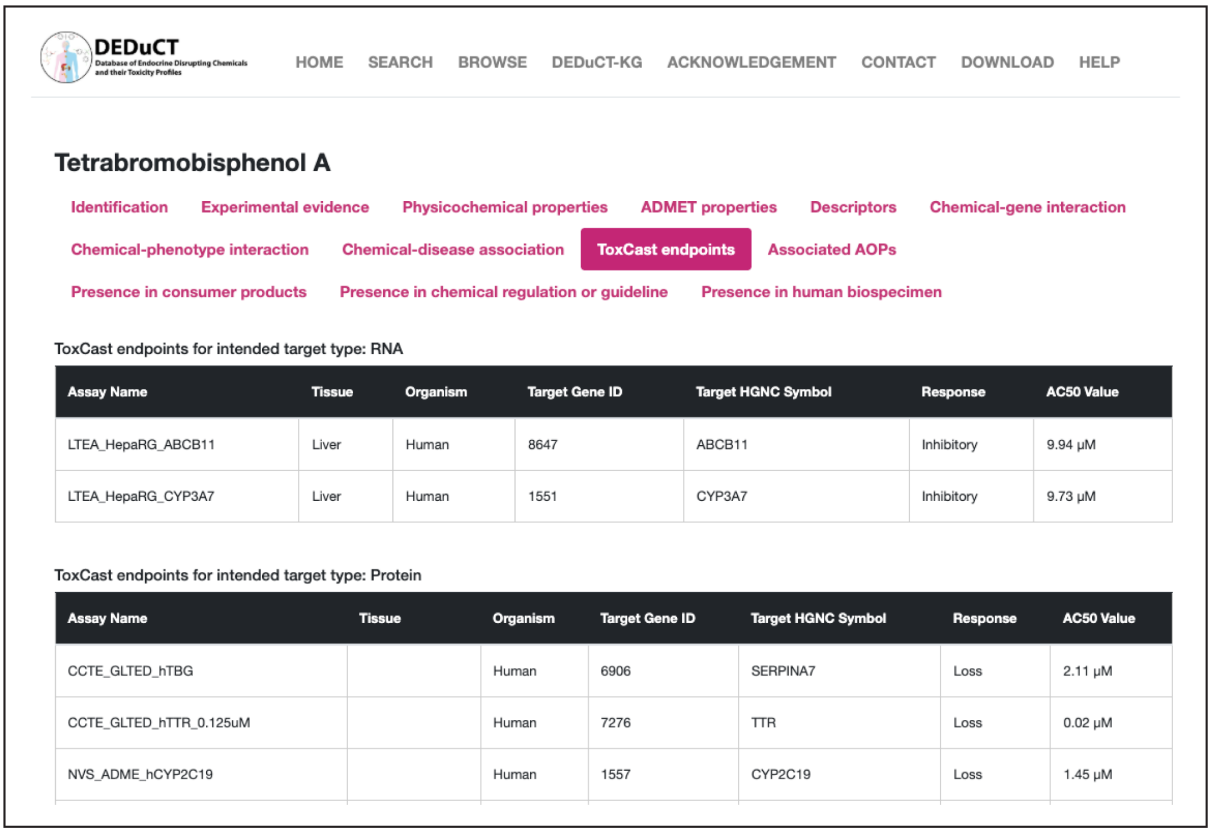


Figure S12. Screenshot from DEDuCTv3.0 webserver of the chemical information page on associated ToxCast endpoints.

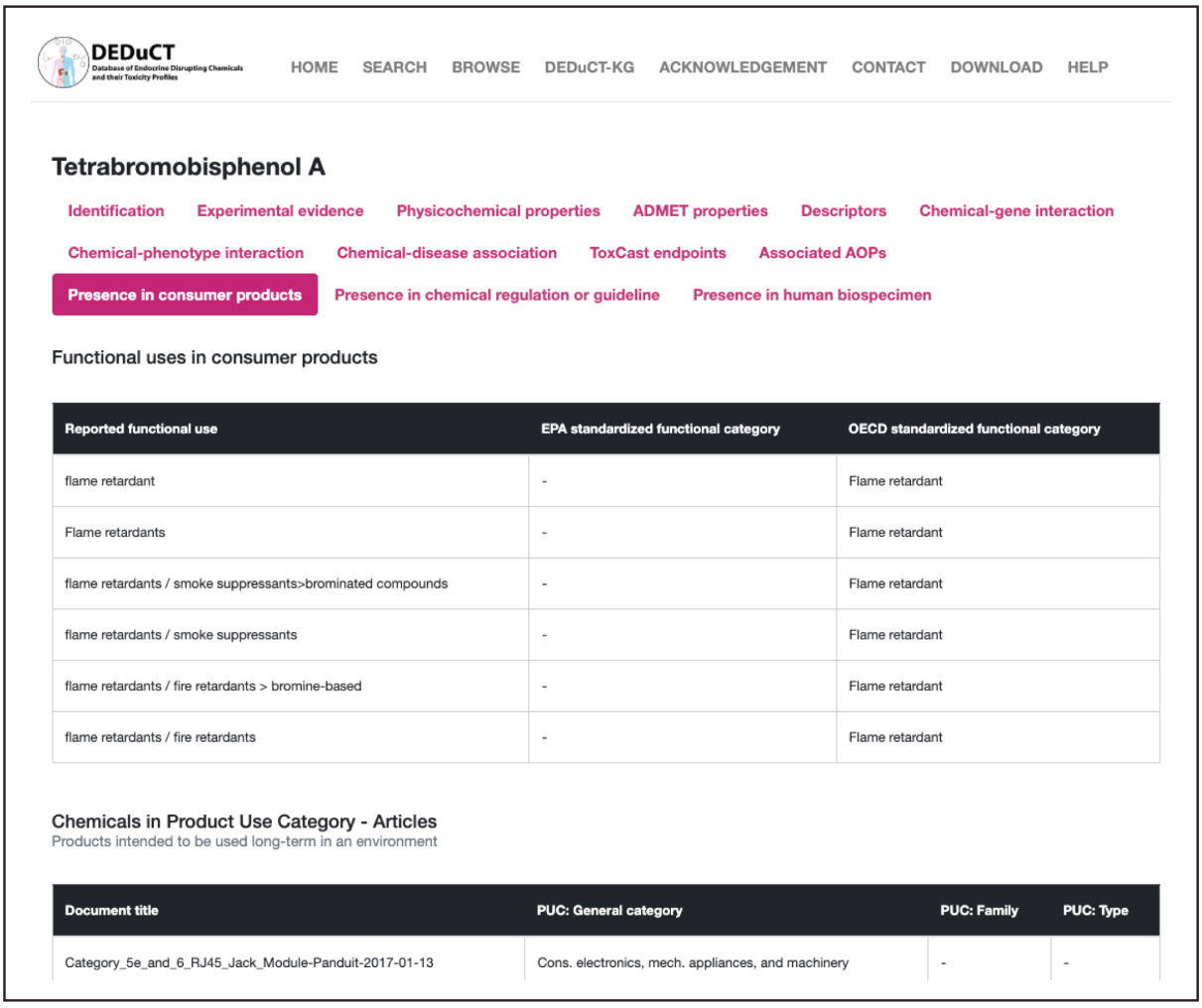


Figure S13. Screenshot from DEDuCTv3.0 webserver of the chemical information page on associated data from Chemical and Products Database (CPDat).

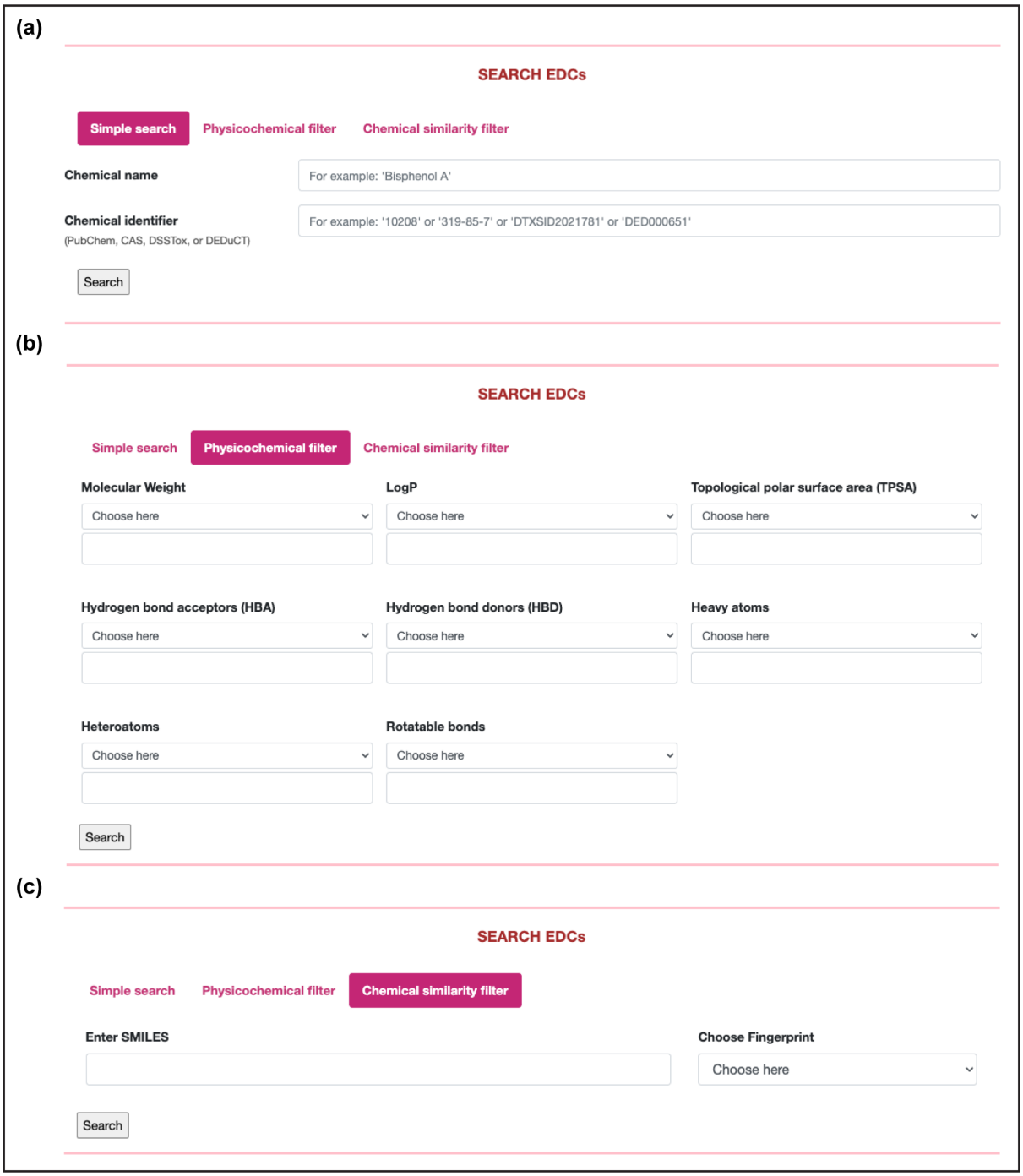


Figure S14. Screenshots from DEDuCTv3.0 webserver for the different search options. (**a**) Search based on chemical information such as names, synonyms, PubChem identifier, CASRN, DSSTox identifier, and DEDuCT identifier. (**b**) Search based on physicochemical properties of chemicals. (**c**) Search based on structural similarity.